Linear Regression Model Training and Prediction

Linear Regression is imported from the Scikit-learn library and instantiated to build a predictive machine learning model. The model is trained using the prepared training dataset (X_train and y_train), allowing it to learn the relationship between the selected input features and the Human Development Index (HDI). After training, the model can predict HDI values for new input data and evaluate its performance using the testing dataset.

Testing the Model

After training, sample input values are passed to the **predict()** method to estimate the HDI score. The predicted output is then compared with the actual values from the testing dataset to assess the model's performance.

```
#testing with few values
y_pred=reg.predict([[13,72.0,5.2,3341.0,14.4]])
print(y_pred)
```

```
[[0.59410218]]
```

Testing the test field form the split:

```
#y_test Values
y_test
```

```
array([0.727, 0.64 , 0.788, 0.579, 0.689, 0.722, 0.682, 0.776])
```